Mohammad Haider

Software engineer building reliable production data and platform systems.

hi@mhaider.dev | mhaider.dev | linkedin.com/in/haidmoham | github.com/haidmoham

SUMMARY

Production software engineer for data systems, internal platforms, and developer tools. I care about software that is correct, easy to reason about, and maintainable by the next person.

SELECTED PROJECTS

C-1N + Robotics Test Bench | Python, MuJoCo, Validation | mhaider.dev/c1n *Jul 2026–Present*

- Built a focused experiment harness to isolate sag in C-1N's original two-joint legs; used the result to add an orthogonal coxa degree of freedom that enabled outward bracing under load.
- Implemented a reproducible 10-second standing acceptance test with checks for six contacts, at least 0.23 m of COM-projection margin, at least 0.45 m of torso height, and numerically zero torso angular speed.
- Built a 33-case disturbance regression suite using 200 ms horizontal force pulses across eight bearings; logged contact, support margin, and torso response, retaining 1x-body-weight failures as an explicit recovery boundary.

PROFESSIONAL EXPERIENCE

University of Virginia | IT Analyst Intermediate, BI Developer | Charlottesville, VA *May 2025–Mar 2026*

- Led analytics projects from stakeholder interviews through deployment and maintenance; translated loosely defined operational questions and cross-organizational data needs into SQL-backed Tableau dashboards.
- Built and redesigned the ReUSE Store's environmental and economic impact dashboard as workflows evolved; added decision-oriented views and supported a product that recorded at least 400 views.
- Collaborated on telemetry-informed performance work that improved organization-wide dashboard load times by approximately 20%.
- Reconciled conflicting purchase-order states across financial systems, tracing discrepancies back to their source data and building a dashboard that surfaced deltas and quantified their financial impact.

Microsoft | Software Engineer I | Redmond, WA *Jul 2021–Jul 2024*

- Owned an end-to-end migration from SSAS Multidimensional to Tabular; defined parity checks, monitored four weeks of pipeline behavior, and kept the legacy service available until the replacement met cutover criteria.
- Wrote and deployed Azure Data Factory verification scripts that monitored daily pipeline runs and checked data consistency between legacy and Tabular services.
- Validated downstream financial-reporting workflows that showed up to 200x faster filtering and approximately 4x faster Excel loading.
- Built a full-stack C#/.NET prototype that replaced a research-heavy manual configuration process with a guided workflow; explained choices, generated configuration files, and validated output.

Microsoft | Software Engineering Intern | Redmond, WA *Summers 2017, 2018*

- Built a C# service that determined each OneDrive user's most recently used items for a product UI feature and validated it in development.

TECHNICAL SKILLS

Languages: C#, Python, SQL, R

Production systems: C#/.NET, Azure Data Factory, Azure Synapse, Kusto (KQL)

Developer workflow: Git, GitHub Actions, Codex, Claude Code, Hermes; additional: MuJoCo, NumPy, pandas, Matplotlib, Jupyter

EDUCATION

Virginia Tech | B.S. in Statistics *2021*

Coursework: Bayesian Statistics, Experimental Design, Statistical Computing, Mathematical Modeling, Combinatorics, and Proofs